LOAN SHARK

AI-Powered Loan Processing

9 Agents | One Decision | Zero Shortcuts

Band of Agents Hackathon 2026

Track 3: Regulated & High-Stakes Workflows

Team: TrenCoders

9	3	94	100%
Specialized AI Agents	Decision Outcomes	Preflight Checks	Human Sign-off

This document provides a complete technical reference for Loan Shark — a 9-agent AI loan processing pipeline built on Band SDK, LangGraph, and Groq.

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1. Executive Summary

Loan Shark is a production-grade, multi-agent AI system that automates the end-to-end processing of loan applications through a sequential pipeline of 9 specialized AI agents. Built for the Band of Agents Hackathon 2026 (Track 3: Regulated & High-Stakes Workflows), the system demonstrates how autonomous agents can handle complex, regulated financial workflows while maintaining full compliance and mandatory human oversight.

Metric	Value
Number of AI Agents	9 specialized, independent agents
LLM Model	Groq llama-3.3-70b-versatile
Agent Runtime	LangGraph + Band SDK
Coordination Protocol	Band @mention real-time messaging
Decision Outcomes	APPROVE / DENY / COUNTER_OFFER
Compliance Framework	RBI Lending Guidelines (India)
Human Oversight	Mandatory at every final decision
Audit Trail	Band room conversation history (immutable)
Frontend	Streamlit web application
System Validation Checks	94 preflight assertions across 10 categories
Language	Python 3.11+
Package Manager	uv (Astral)

Key Innovation

The system uses Band's real-time messaging as BOTH the coordination layer AND the compliance audit trail. Every agent handoff, every piece of enriched data, and every decision flag is recorded in the Band room as an immutable, timestamped message. This dual-purpose architecture means the legally required audit trail emerges naturally from the system's operation — no separate logging infrastructure needed.

2. The Problem

The traditional loan processing industry suffers from systemic inefficiencies that cost financial institutions billions annually and frustrate applicants with opaque, slow, and inconsistent experiences.

Pain Point	Severity	Description
Slow Processing Times	3-14 Days	Applications bounce between fraud, credit, compliance, and legal departments — each with its own queue, tooling, and communication style. Applicants wait. Interest accrues. Competitors win.
Data Loss at Handoffs	Critical Risk	When a loan officer passes an application to a fraud analyst to a compliance officer, context is routinely lost. Notes are informal, context is verbal. By the time a decision is made, the compliance team is working with an incomplete picture.
Inconsistent Decision-Making	Regulatory Risk	Different loan officers apply different thresholds. Two identical applications may get different outcomes depending on who processes them, exposing institutions to fair lending violations.
Audit Trail Fragmentation	Compliance Gap	Reconstructing a decision for regulatory review requires pulling emails, system logs, manual notes, and database queries — an expensive, error-prone process that often fails to produce a coherent narrative.
Human Bottlenecks	Scalability Limit	Every step requires a human handoff. Scaling loan volume means scaling headcount proportionally. There is no path to exponential throughput improvement without fundamentally redesigning the process.

3. The Solution

Loan Shark replaces the fragmented, manual handoff chain with a structured 9-agent AI pipeline. Each agent has a single, well-defined responsibility, receives structured input, performs its analysis using Groq's llama-3.3-70b-versatile LLM, and passes an enriched structured payload to the next agent via Band's messaging protocol.

Key Differentiators

- **Speed:** What takes 3-14 days in a traditional process completes in minutes. AI agents don't have queues, meetings, or lunch breaks.
- **Consistency:** Every application is evaluated against the exact same criteria in the exact same order. No human variance, no implicit bias.
- **Complete Context:** Each downstream agent receives everything its predecessors computed — enriched, structured, and validated. No information loss at handoffs.
- **Built-in Compliance:** A dedicated Compliance Agent applies RBI regulatory requirements as a hard gate. No approved application bypasses regulatory review.
- **Immutable Audit Trail:** The Band room history IS the audit trail. Every handoff, every flag, every decision note — timestamped and immutable.
- **Human Authority:** AI is advisory. The final decision always requires a human loan officer's signature. The system augments human judgment, not replaces it.

4. System Architecture

4.1 High-Level Overview

Loan Shark is composed of three primary layers: the Presentation Layer (Streamlit), the Agent Layer (9 independent Band agents), and the Coordination Layer (Band platform). These layers communicate exclusively through Band's messaging infrastructure.

Layer	Component	Responsibility
Presentation	Streamlit UI	Loan application form, real-time pipeline status, human gate interface
Coordination	Band Platform	Message routing, @mention delivery, room history, WebSocket connections
Agent (x9)	Python + LangGraph	Domain-specific analysis, LLM inference via Groq, structured output
LLM Backend	Groq API	Fast inference for all 9 agents using llama-3.3-70b-versatile

4.2 Pipeline Architecture

The pipeline is a strict sequential chain. Each agent can only trigger the next; there is no branching, no parallelism, and no backtracking. This deliberate design ensures a consistent, auditable, and reproducible processing sequence.

FULL PIPELINE FLOW DIAGRAM

```
Applicant fills form (Streamlit UI)
      |
      v  POST via Band REST API
[1] INTAKE AGENT           Validates & structures application
      |  LOAN_APPLICATION: @DocumentAgent
      v
[2] DOCUMENT AGENT         Doc completeness & consistency
      |  DOC_VERIFICATION: @CreditAgent
      v
[3] CREDIT AGENT           Credit profile & grade analysis
      |  CREDIT_ANALYSIS: @FraudAgent
      v
[4] FRAUD AGENT            Fraud signals & identity checks
      |  FRAUD_REPORT: @RiskAgent
      v
[5] RISK AGENT             DTI ratio & financial risk scoring
      |  RISK_ASSESSMENT: @ComplianceAgent
      v
[6] COMPLIANCE AGENT       RBI regulatory compliance check
      |  COMPLIANCE_CHECK: @DecisionAgent
      v
[7] DECISION AGENT         APPROVE / DENY / COUNTER_OFFER
      |  LOAN_DECISION_READY: @PricingAgent
      v
[8] PRICING AGENT          Interest rate, EMI, fees, terms
      |  PRICING_TERMS: @CommunicationAgent
      v
[9] COMMUNICATION AGENT    Formal sanction / rejection letter
      |  FORMAL_LETTER_READY:
      v
HUMAN LOAN OFFICER         Reviews AI recommendation & signs off
```

4.3 Communication Model

All inter-agent communication occurs exclusively through Band's real-time messaging. Each agent is a persistent WebSocket connection to the Band platform. When an agent receives a message containing its trigger keyword, it wakes up, processes the payload using LangGraph + Groq, generates a response, and posts the next message with an @mention of the next agent. There is no shared memory, no message queue, no service bus, no database. The Band room is the single source of truth.

4.4 Data Enrichment Chain

Each agent receives the previous agent's complete JSON payload and extends it with its own computed fields before passing it forward. By Agent 9, the payload contains every field computed by every previous agent — a complete, structured application history in a single document.

After Agent	New Fields Added to Payload
[1] Intake	application_id, validated schema, timestamp, currency

After Agent	New Fields Added to Payload
[2] Document	doc_verdict, doc_flags[], doc_completeness_score
[3] Credit	credit_grade, credit_tier, max_lti, credit_behavior_notes
[4] Fraud	fraud_risk_level, fraud_flags[], fraud_confidence
[5] Risk	risk_score, dti_before, dti_after, employment_risk, collateral_coverage + all prior fields
[6] Compliance	compliance_verdict, violations[], regulatory_notes
[7] Decision	recommendation, approved_amount, confidence, decision_basis, compliance_notes
[8] Pricing	interest_rate, processing_fee, emi, prepayment_terms, insurance_premium
[9] Communication	letter_type, letter_body (full formatted letter text)

5. The 9-Agent Pipeline

Each agent is an independent Python process. It connects to the Band platform via WebSocket, listens for its trigger keyword, performs its domain-specific analysis using LangGraph and Groq, then posts its structured output back to the Band room with an @mention of the next agent. All 9 run simultaneously — each sleeping until its keyword appears.

Agent 1 of 9 — Intake Agent

Trigger: NEW_LOAN_APPLICATION: **Output:** LOAN_APPLICATION: @DocumentAgent

The gatekeeper of the pipeline. Receives raw, unstructured application data submitted via the Streamlit UI and performs the first line of validation and structuring. If validation fails, it fires INTAKE_ERROR: and halts the pipeline immediately, preventing corrupt data from propagating through the remaining 8 agents.

Responsibilities:

- Validates all required fields are present (name, age, income, loan amount, purpose, tenure)
- Sanity-checks values: age 18-70, income > 0, loan amount within plausible bounds
- Detects logical errors (e.g. claiming 20 years employment at age 22)
- Generates a unique application_id (format: APP-{timestamp})
- Normalizes currency representation and numeric fields
- Formats raw text into a structured LoanApplication JSON schema
- On failure: fires INTAKE_ERROR: with detailed error code and halts pipeline

Agent 2 of 9 — Document Verification Agent

Trigger: LOAN_APPLICATION: **Output:** DOC_VERIFICATION: @CreditAgent

Checks document completeness and internal consistency without connecting to any external database. Its analysis is entirely logical — cross-referencing claimed employment type against expected documentation requirements.

Document Requirements by Employment Type:

Employment Type	Required Documents
Salaried	Salary slips (3 months), employee ID, bank statements (6 months), employer letter
Self-Employed	ITR (2 years), business registration certificate, GST registration, bank statements
Business Owner	Company registration, audited financials (2 years), ITR, bank statements
Unemployed	Alternate income proof (rental income, investments), bank statements

Output fields: doc_verdict (COMPLETE / PARTIAL / INSUFFICIENT), doc_completeness_score (0-100), doc_flags[] with specific issues identified.

Agent 3 of 9 — Credit Analysis Agent

Trigger: DOC_VERIFICATION: **Output:** CREDIT_ANALYSIS: @FraudAgent

Interprets the raw CIBIL/credit score into a full credit profile. Determines creditworthiness tier and the maximum exposure the institution should be willing to take for this applicant.

Credit Grade Mapping:

Score Range	Grade	Risk Tier	Max Loan-to-Income
750 - 900	**A+**	Prime	Up to 5x monthly income
700 - 749	**A**	Near-Prime	Up to 4x monthly income
650 - 699	**B**	Subprime	Up to 3x monthly income
600 - 649	**C**	High Risk	Up to 2x monthly income
550 - 599	**D**	Very High Risk	Up to 1.5x monthly income
Below 550	**F**	Decline Territory	Hard decline recommended
Not Provided	**N/A**	First-Time Borrower	Max 1x monthly income

Agent 4 of 9 — Fraud Detection Agent

Trigger: CREDIT_ANALYSIS: **Output:** FRAUD_REPORT: @RiskAgent

The pattern recognition specialist. Analyzes the complete application for fraud signals, inconsistencies, and high-risk behavioral patterns that suggest the application may be fraudulent, synthetic, or part of a loan stacking scheme.

Fraud Signal Categories Checked:

Category	What the Agent Checks
Income Anomalies	Income claims disproportionate to employment type, age, or sector norms. Self-employed applicants claiming corporate-level income with no business registration details.
Identity Inconsistencies	Mismatched employment tenure vs. age. Claimed education level inconsistent with income. Employer-address combinations geographically implausible.
Loan Stacking Indicators	Existing debt levels suggesting multiple concurrent loan applications. DTI at intake that implies prior undisclosed credit facilities.
Synthetic Identity Signals	No credit history combined with high income claims. Perfect documentation on an otherwise high-risk profile. Unusually long tenure at unknown companies.
Application Velocity Patterns	Loan amount at suspiciously round numbers. Purpose codes inconsistent with the amount/tenure requested.

Output: fraud_risk_level (LOW / MEDIUM / HIGH / CRITICAL) and fraud_flags[] array with specific flag codes and human-readable reasoning.

Agent 5 of 9 — Risk Assessment Agent

Trigger: FRAUD_REPORT: **Output:** RISK_ASSESSMENT: @ComplianceAgent

The financial modeler and data aggregator. Performs quantitative risk scoring using DTI analysis, employment stability assessment, and collateral valuation. Also carries forward ALL structured fields from Agents 1-4 so downstream agents have complete context without back-referencing earlier messages.

Calculations Performed:

- DTI Before Loan: (existing_monthly_debt / monthly_income) x 100
- DTI After Loan: ((existing_monthly_debt + estimated_emi) / monthly_income) x 100
- Employment Stability Score: Weighted (tenure, employment type, employer size)
- Collateral Coverage Ratio: Estimated collateral value / loan amount requested
- Composite Risk Score (0-100): Weighted avg of credit grade + DTI + fraud level + employment + doc completeness
- Risk Category: LOW (0-30) / MEDIUM (31-55) / HIGH (56-75) / CRITICAL (76-100)

Agent 6 of 9 — Compliance Agent

Trigger: RISK_ASSESSMENT: **Output:** COMPLIANCE_CHECK: @DecisionAgent

The regulatory guardrail. Validates the application against RBI lending guidelines and fair lending principles. Its verdict can block an application regardless of positive credit or risk scores — it is the one agent whose hard block cannot be overridden by the AI pipeline.

RBI Guidelines Applied:

Guideline	Rule Applied
DTI Cap	Post-loan DTI must not exceed 50% for personal/vehicle loans, 60% for home loans with collateral
KYC Requirements	All mandatory identity and income documents must be present per employment type
Sector Exposure Limits	Loan purpose cross-checked against RBI sector exposure guidelines
Minimum Income Floor	Monthly income must be sufficient to cover EMI with a minimum 20% buffer remaining
Age-Tenure Alignment	Loan tenure must not extend beyond retirement age (65) for salaried employees
Fair Lending Check	Flags if any combination of factors could constitute discriminatory lending
Maximum LTV Ratio	For secured loans, loan amount must not exceed 80% of estimated collateral value

Agent 7 of 9 — Decision Agent

Trigger: COMPLIANCE_CHECK: **Output:** LOAN_DECISION_READY: @PricingAgent

Applies a deterministic decision matrix to the full context from all 6 upstream agents. Produces the official lending recommendation with full compliance documentation. Every decision is accompanied by a compliance audit note citing the specific rule or factor combination that triggered it.

Decision Matrix:

Condition	Decision
compliance_verdict == NON_COMPLIANT	**DENY (hard block - regulatory)**
fraud_risk_level == CRITICAL	**DENY (hard block - fraud)**
credit_grade == F	**DENY (hard block - creditworthiness)**
DTI after loan > 70%	**DENY (hard block - affordability)**
All hard blocks clear AND risk_score <= 40	**APPROVE**
All clear AND risk_score 41-65	**COUNTER_OFFER (reduced amount/tenure)**
Any soft flag AND risk_score > 65	**DENY (with improvement guidance)**

Agent 8 of 9 — Pricing Agent

Trigger: LOAN_DECISION_READY: **Output:** PRICING_TERMS: @CommunicationAgent

The actuary. Calculates exact financial terms within the parameters established by the Decision Agent, personalizing the offer to the specific risk profile of the applicant.

Pricing Calculations:

- Base rate: RBI repo rate + credit grade spread
- Risk premium: Added based on risk_score band
- Final interest rate: Base rate + credit spread + risk premium
- Processing fee: 0.5-2% of loan amount based on loan type and risk
- Monthly EMI: Standard amortization formula using principal, rate, and tenure
- Insurance premium: If mandatory, based on loan amount and tenure
- Prepayment penalty structure: 0-2% of outstanding principal based on remaining tenure
- COUNTER_OFFER path: All terms recalculated for the modified amount and tenure

Agent 9 of 9 — Communication Agent

Trigger: PRICING_TERMS: **Output:** FORMAL_LETTER_READY: -> Human Gate

The writer. Drafts the formal, legally-worded correspondence that will be sent to the applicant. Critically, this agent does NOT dispatch letters autonomously — the draft goes to the Human Gate for loan officer review and signature.

Letter Type	Contents
Loan Sanction Letter (APPROVE / COUNTER_OFFER)	Formal header (Loan Shark Financial Services), applicant salutation, congratulatory opening, all approved financial terms (amount / tenure / rate / EMI / fee), key conditions (3-4 bullet points), next steps for document submission, legal disclaimer, officer signature block
Regret Letter (DENY)	Professional empathetic tone, specific denial reasons in regulatory language, guidance on improving eligibility (credit score improvement, debt reduction), timeline for re-application, grievance redressal information

6. Message Protocol & Data Schema

6.1 Message Format

Each agent communicates by posting a structured message to the Band room. The message has two parts: a plain-text header with the message tag and @mention, and a JSON payload embedded in a markdown code fence.

EXAMPLE: Intake Agent Output Message

````
LOAN_APPLICATION: Application ID: APP-123456
@DocumentAgent please verify this application.

```json
{
  "message_type": "LOAN_APPLICATION",
  "application_id": "APP-123456",
  "applicant_name": "Priya Sharma",
  "applicant_age": 32,
  "monthly_income": 120000.0,
  "currency": "INR",
  "employment_type": "salaried",
  "employer_name": "Infosys Ltd",
  "years_employed": 5.0,
  "loan_amount_requested": 800000.0,
  "loan_purpose": "home",
  "loan_tenure_months": 120,
  "existing_debt_monthly": 8000.0,
  "credit_score": 768,
  "collateral_offered": "Residential property in Bengaluru"
}
```
````

6.2 Full Message Tag Reference

Message Tag	Sender	Receiver	Key Fields Added
NEW_LOAN_APPLICATION:	Streamlit UI	IntakeAgent	Raw form data as plain text
LOAN_APPLICATION:	IntakeAgent	DocumentAgent	application_id, validated schema
DOC_VERIFICATION:	DocumentAgent	CreditAgent	doc_verdict, doc_flags[], doc_completeness_score
CREDIT_ANALYSIS:	CreditAgent	FraudAgent	credit_grade, credit_tier, max_lti
FRAUD_REPORT:	FraudAgent	RiskAgent	fraud_risk_level, fraud_flags[], fraud_confidence

Message Tag	Sender	Receiver	Key Fields Added
RISK_ASSESSMENT:	RiskAgent	ComplianceAgent	risk_score, dti_before, dti_after + all prior
COMPLIANCE_CHECK:	ComplianceAgent	DecisionAgent	compliance_verdict, violations[]
LOAN_DECISION_READY:	DecisionAgent	PricingAgent	recommendation, approved_amount, confidence
PRICING_TERMS:	PricingAgent	CommunicationAgent	interest_rate, emi, processing_fee
FORMAL_LETTER_READY:	CommunicationAgent	Human Gate	letter_body, letter_type
INTAKE_ERROR:	IntakeAgent	Human / System	error_code, error_detail (HALTS pipeline)

7. Technology Stack

7.1 Band SDK

Band SDK is the core coordination infrastructure. It provides the real-time WebSocket connections that allow agents to listen for and publish messages in a shared room. The @mention protocol is the sole mechanism by which agents hand off work to each other.

Component	Purpose
band.Agent	Base class for all 9 agents — handles WebSocket lifecycle, reconnection, and message dispatch
band.adapters.LangGraphAdapter	Bridges Band message events to LangGraph's agent graph execution model
band.config.load_agent_config	Loads agent UUID and API key from agent_config.yaml
WebSocket URL	wss://app.band.ai/api/v1/socket/websocket
REST API URL	https://app.band.ai/ — used by Streamlit to post the initial application

7.2 LangGraph

LangGraph provides the agent execution runtime. Each agent is implemented as a LangGraph StateGraph with nodes for input parsing, LLM inference, output validation, and message posting. The InMemorySaver checkpointer maintains state within a session.

7.3 Groq LLM

All 9 agents use Groq's llama-3.3-70b-versatile model via LangChain's ChatOpenAI interface (Groq is OpenAI-API-compatible). Groq's inference speed is critical — it enables the full 9-agent pipeline to complete in under 2 minutes.

LLM CONFIGURATION (same across all 9 agents)

```
llm = ChatOpenAI(
    model="llama-3.3-70b-versatile",
    openai_api_key=os.getenv("GROQ_API_KEY"),
    openai_api_base="https://api.groq.com/openai/v1",
    temperature=0.1,   # Low: consistent, structured outputs
    max_tokens=2048,
)
```

7.4 Streamlit Frontend

The Streamlit application serves as both the loan application interface and the real-time pipeline monitoring dashboard. Uses Streamlit's session_state to track pipeline status, agent messages, and decision data across reruns.

7.5 Package Management — uv

uv provides deterministic dependency resolution via uv.lock, fast installation, and the 'uv run' prefix for all Python script execution.

Package	Version	Purpose
band-sdk	latest	Agent framework + WebSocket + @mention protocol
langchain-openai	latest	OpenAI-compatible LLM client interface to Groq
langgraph	latest	Agent state machine and graph execution runtime
streamlit	latest	Web UI framework
python-dotenv	latest	Environment variable loading from .env
pyyaml	latest	agent_config.yaml parsing
requests	latest	Band REST API calls from Streamlit
reportlab	4.5.1	PDF documentation generation

8. Compliance & Regulatory Framework

Compliance is a first-class architectural concern in Loan Shark. The system enforces regulatory requirements at multiple levels, with different enforcement strengths for different rule types.

8.1 Hard Blocks vs Soft Flags

Type	Effect	Override?	Examples
Hard Block	Automatic DENY	Human only	Regulatory violation, CRITICAL fraud level, Grade F credit, DTI > 70%
Soft Flag	Reduces approval probability	Decision Agent	MEDIUM fraud risk, PARTIAL documents, HIGH risk score (56-75)
Warning	Logged in audit trail	Automatic	Unusual income claim, first-time borrower, missing collateral on borderline loan

8.2 RBI Guidelines Implemented

DTI Threshold Enforcement

The Compliance Agent applies maximum Debt-to-Income ratios: 50% for unsecured loans (personal, vehicle, education), 60% for secured home loans. Applications breaching these thresholds receive a hard block regardless of credit score.

KYC Verification Requirements

Employment-type-specific document requirements are enforced. Salaried employees must provide salary slips and employment verification; self-employed must have ITR and business registration; all applicants require valid government photo ID.

Age-Tenure Alignment Rule

For salaried applicants, the loan tenure must not extend beyond the standard retirement age of 65. A 55-year-old cannot receive a 15-year salaried home loan product.

Minimum Income Requirements

Post-EMI take-home income must not fall below 40% of gross monthly income. Applications where the proposed EMI would breach this threshold are blocked.

Sector Exposure Guidelines

The compliance agent checks whether the loan purpose falls under any RBI sector-specific exposure caps (e.g. real estate, commercial vehicle). High-exposure sectors trigger additional scrutiny flags.

8.3 Fair Lending Checks

The Compliance Agent includes a fair lending verification step. It checks that the combination of factors used in the recommendation could not be construed as indirectly discriminatory under Indian fair lending regulations. Decisions must be defensible on purely financial grounds. Any decision that appears to penalize an applicant on non-financial grounds is flagged with a FAIR_LENDING_REVIEW code.

9. Human-in-the-Loop Design

Human oversight is the non-negotiable design principle of Loan Shark. The AI pipeline is an advisor and accelerator, not a decision-maker. No loan can be approved, denied, or disbursed without a human loan officer's explicit authorization.

The Human Gate — What the Loan Officer Sees

When the Communication Agent posts FORMAL_LETTER_READY:, the Streamlit application enters the Human Gate state. The loan officer is presented with:

- Decision summary card: Recommendation badge (APPROVE / DENY / COUNTER_OFFER), application ID, risk category, and AI confidence score
- Key financial metrics: Approved loan amount, tenure, estimated monthly EMI
- Counter-offer terms (if applicable): Modified amount, rate band, and reasoning
- Denial reasons (if applicable): Specific flags that triggered denial in plain language
- Full formal letter preview: Complete, scroll-able bank letter draft as formatted by the Communication Agent
- Compliance & audit trail: Full compliance notes from Agent 6 in an expandable section
- Two action buttons: 'Approve & Finalize' (primary) and 'Reject / Override' (secondary)

Override Authority

The human loan officer retains full authority to override any AI recommendation. Clicking 'Reject / Override' on an APPROVE recommendation is fully supported. All overrides are logged in the agent message feed with timestamp and action type. The institution's credit policy may require senior officer sign-off for overrides of hard-block decisions.

Audit Trail Completeness

The Band room history provides a complete, timestamped audit trail: every agent's analysis, every flag raised, the exact recommendation with its basis, and the human officer's final action. This is sufficient for RBI examination, internal audit, and any applicant grievance resolution.

10. Frontend — Streamlit Application

The Streamlit application (app.py) is a single-page application with a two-column layout: application form on the left, pipeline tracker and agent feed on the right.

Application Form Fields

Field	Type	Validation
Full Name	Text input	Required, non-empty
Age	Number (18-70)	Must be between 18 and 70
Credit Score	Number (0-900)	0 = unknown; maps to first-time borrower path
Currency	Selectbox	INR / USD
Employment Type	Selectbox	salaried / self_employed / business_owner / unemployed
Employer Name	Text input	Required for salaried/self-employed
Monthly Income	Number > 0	Must be positive
Years at Job	Number (0-40)	Cross-checked against applicant age
Loan Amount	Number > 10,000	Minimum loan threshold enforced
Loan Purpose	Selectbox	home / vehicle / education / personal / business
Tenure	Slider (6-360 mo)	Step size: 6 months
Existing Debt/mo	Number >= 0	Used in DTI calculation by Risk Agent
Collateral	Text input	Optional but affects risk score and terms

Quick Demo Scenarios

Three pre-built demo scenarios allow instant demonstration without manual data entry. Clicking a demo button pre-fills all form fields with a carefully designed test case.

Button	Applicant	Profile	Expected Outcome
Good Applicant	Priya Sharma, 32	Salaried, Rs.1.2L/mo, CIBIL 768, home loan, collateral	**APPROVE**
Borderline	Arjun Mehta, 28	Self-employed, Rs.55K/mo, CIBIL 648, vehicle loan, no collateral	**COUNTER_OFFER**
High Risk	Ravi Kumar, 45	Unemployed, Rs.30K/mo, no credit score, personal loan	**DENY**

Pipeline Tracker

The right column displays a 9-card vertical pipeline tracker. Each card shows the agent name and current status, updating dynamically as messages are processed: Waiting (grey) -> Processing (cyan, active border glow) -> Complete (green). The tracker uses detect_message_stage() to classify each pasted message.

Human Gate Interface

When FORMAL_LETTER_READY: is detected, the pipeline tracker enters awaiting_approval state. The human gate section renders below the agent feed with: decision summary, financial metrics (loan amount, tenure, EMI), formal letter preview rendered in Georgia serif font in a scrollable container, compliance audit trail, and the two action buttons.

11. Setup & Deployment Guide

Prerequisites

Requirement	Version	Notes
Python	3.11+	3.13.x recommended. Must be installed and in PATH.
uv	latest	pip install uv OR winget install astral-sh.uv
Groq API Key	Free	Register at console.groq.com — free tier available
Band Account	Pro	app.band.ai — use promo code BANDHACK26 for free Pro access
Git	Any	For cloning the repository

Step-by-Step Installation

Step 1 — Clone the repository

```
git clone https://github.com/Ares19v/Loan_Shark.git
cd Loan_Shark
```

Step 2 — Install all dependencies

```
uv sync
```

Step 3 — Configure environment

```
copy .env.example .env
# Edit .env and fill in:
#   GROQ_API_KEY=your-key-here
#   BAND_ROOM_ID=your-room-id-here
```

Step 4 — Run preflight check

```
uv run python preflight.py
# Fix any FAIL items before proceeding
```

Step 5 — Create Band agents

```
# On app.band.ai:
# 1. Create 9 agents: IntakeAgent, DocumentAgent, CreditAgent,
#    FraudAgent, RiskAgent, ComplianceAgent, DecisionAgent,
#    PricingAgent, CommunicationAgent
# 2. Create one room named 'LoanShark' and add all 9 agents
# 3. Copy each agent's UUID and API key into agent_config.yaml
```

Step 6 — Launch all 9 agents (one command)

```
uv run python run_all.py
```

Step 7 — Launch the Streamlit UI

```
uv run streamlit run app.py
# Opens at http://localhost:8501
```

12. Quality Assurance — Preflight Check

preflight.py is a 94-assertion system validation script inspired by aviation preflight checklists. It validates every component of the system before launch and exits with code 0 (all clear) or code 1 (failures found).

Category	Checks	What It Validates
1. Python & Runtime	9	Python >= 3.11, all 7 required packages importable
2. File Structure	20	All 20 required project files exist on disk
3. Environment Vars	4	.env populated, no placeholder values remaining
4. Agent Config	10	agent_config.yaml valid YAML, all 9 agents have UUID and API key
5. Python Syntax	13	AST-parse all 13 Python source files — zero syntax errors
6. Pipeline Chain	9	Each agent file has correct trigger keyword, @mention, and output tag
7. Streamlit App	16	All 9 message types detected, all 9 stages displayed, demo scenarios present
8. Branding	9	No stale 'LoanPilot' references in any file across the codebase
9. API Connectivity	2	Live Groq API call to llama-3.3-70b-versatile succeeds; Band platform reachable
10. SDK Imports	5	Band SDK, LangGraph, LangChain all importable without errors
TOTAL	94+	Complete critical path validation — run after every significant code change

PREFLIGHT OUTPUT EXAMPLE

```
uv run python preflight.py

# Clean system output:
# PREFLIGHT SUMMARY
# Passed  : 85
# Warnings: 9    <- Band credential placeholders (expected pre-setup)
# Failed  : 0
# -> CLEAR for launch after filling Band credentials
```

13. Hackathon Track Alignment

Loan Shark was designed from the ground up to address every judging criterion of Track 3: Regulated & High-Stakes Workflows.

Track 3 Requirement	How Loan Shark Addresses It
Human oversight at critical decision points	Mandatory human gate before any letter is dispatched. Loan officer sees full decision context and must actively approve or override. AI cannot finalize independently.
Complete audit trail for regulatory review	Band room conversation history is the full, timestamped, immutable audit trail. Every agent output with every computed field is preserved in sequence.
Compliance with domain regulations	Dedicated Compliance Agent applies RBI lending guidelines as a hard gate. Violations issue hard blocks that override positive risk/credit scores.
Structured, reproducible decision-making	Deterministic decision matrix in the Decision Agent. Same input conditions always produce the same decision path. Full explainability.
Multi-agent coordination for complex workflows	9 specialized agents via Band @mention — each with a single, well-defined responsibility and full accountability for its output.
Error handling and graceful degradation	INTAKE_ERROR: halts pipeline on bad data. Each agent validates its own input before processing. Pipeline never proceeds with corrupt data.
Deep integration with Band SDK	Band SDK is the only coordination mechanism. @mention is the sole inter-agent protocol. Band REST API posts the initial trigger. WebSocket maintains persistent agent connections. Remove Band — the pipeline collapses entirely.

14. Future Roadmap

Short Term (Post-Hackathon)

- Auto-polling: Streamlit polls Band REST API every 3 seconds to automatically display agent messages without manual paste
- Document upload: Allow actual PDF document uploads and use OCR to extract and verify document data
- Retry logic: Automatic retry with exponential backoff if any agent's LLM call fails
- Streamlit Cloud deployment: Production deployment with Streamlit secrets for API keys
- Multi-currency support: Full USD / GBP / EUR support with appropriate regulatory framework switching
- Agent timing display: Per-agent processing time shown in the pipeline tracker

Medium Term

- Bank integration API: REST endpoints for integration with existing Core Banking Systems (CBS)
- Applicant portal: Separate applicant-facing interface showing application status
- Batch processing: Queue-based processing for high-volume periods
- A/B testing framework: Compare decision outcomes across different model configurations
- Regulatory report generation: Auto-generate RBI quarterly compliance reports from Band room history

Long Term Vision

Loan Shark's architecture is generalisable beyond loan processing to any regulated, sequential, multi-stakeholder workflow. Insurance claims processing, KYC verification, trade compliance, and regulatory filing workflows all share the same pattern: specialized expertise at each stage, structured data handoffs, compliance gates, and mandatory human sign-off. The Band @mention protocol generalizes to any domain.

Appendix A — Project File Structure

```
Loan_Shark/
|
+-- agents/
|   +-- intake/agent.py          Agent 1: validates & structures raw application
|   +-- document/agent.py        Agent 2: document completeness & consistency
|   +-- credit/agent.py          Agent 3: credit profile & grade analysis
|   +-- fraud/agent.py           Agent 4: fraud signals & identity checks
|   +-- risk/agent.py            Agent 5: DTI, financial risk scoring, data aggregator
|   +-- compliance/agent.py      Agent 6: RBI regulatory compliance
|   +-- decision/agent.py        Agent 7: APPROVE / DENY / COUNTER_OFFER
|   +-- pricing/agent.py         Agent 8: interest rate, EMI, fees, prepayment
|   +-- communication/agent.py   Agent 9: formal sanction/rejection letter
|
+-- schema/
|   +-- messages.py              Shared message tag constants
|
+-- .streamlit/
|   +-- config.toml              Dark theme + server configuration
|
+-- app.py                       Streamlit UI: form + pipeline tracker + human gate
+-- run_all.py                   Single-command launcher for all 9 agents
+-- preflight.py                 94-check system validation script
+-- generate_pdf.py              This PDF document generator
|
+-- agent_config.yaml            Band agent UUIDs and API keys  [GITIGNORED]
+-- .env                         Environment variables          [GITIGNORED]
+-- .env.example                 Environment variable template
|
+-- pyproject.toml               Project metadata and dependencies
+-- requirements.txt             Streamlit Cloud deployment requirements
+-- uv.lock                      Deterministic dependency lock file
+-- README.md                    Project documentation
```

Appendix B — Configuration Reference

.env Variables

Variable	Required	Description
GROQ_API_KEY	Yes	Groq API key from console.groq.com
BAND_REST_URL	Yes	https://app.band.ai/ (do not modify)
BAND_WS_URL	Yes	wss://app.band.ai/api/v1/socket/websocket (do not modify)
BAND_ROOM_ID	Yes*	Room ID from band.ai room URL after creating the room
APP_ENV	No	development or production (default: development)

agent_config.yaml Structure

```
# agent_config.yaml
# DO NOT COMMIT — this file is gitignored

intake:
  agent_id: "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx"
  api_key:  "band_xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx"

document:
  agent_id: "..."
  api_key:  "..."

credit:
  agent_id: "..."
  api_key:  "..."

fraud:
  agent_id: "..."
  api_key:  "..."

# ... (repeat for: risk, compliance, decision, pricing, communication)
```

Appendix C — API Message Reference

Complete Message Tag Glossary

NEW_LOAN_APPLICATION:

Initiating message from Streamlit UI to Intake Agent. Contains raw form data as plain text — not a JSON payload. Parsed by the Intake Agent from the natural language format.

LOAN_APPLICATION:

Intake Agent to Document Agent. Contains fully structured, validated LoanApplication JSON. application_id is assigned here. This is the first properly structured JSON in the chain.

DOC_VERIFICATION:

Document Agent to Credit Agent. Adds doc_verdict (COMPLETE/PARTIAL/INSUFFICIENT), doc_flags[] array of specific issues found, and doc_completeness_score (0-100).

CREDIT_ANALYSIS:

Credit Agent to Fraud Agent. Adds credit_grade (A+/A/B/C/D/F), credit_tier (Prime/Near-Prime/Subprime/etc.), max_lti (maximum loan-to-income ratio), credit_behavior_notes (first-time borrower flag, utilization assessment).

FRAUD_REPORT:

Fraud Agent to Risk Agent. Adds fraud_risk_level (LOW/MEDIUM/HIGH/CRITICAL), fraud_flags[] array with flag codes and human-readable reasoning, fraud_confidence (confidence score of the fraud assessment 0-100).

RISK_ASSESSMENT:

Risk Agent to Compliance Agent. Adds risk_score (0-100), risk_category (LOW/MEDIUM/HIGH/CRITICAL), dti_before, dti_after, employment_risk_score, collateral_coverage_ratio. ALSO re-carries ALL fields from agents 1-4 to ensure Compliance Agent has complete context.

COMPLIANCE_CHECK:

Compliance Agent to Decision Agent. Adds compliance_verdict (COMPLIANT/NON_COMPLIANT/CONDITIONAL), violations[] array listing any breached guidelines with specific RBI regulatory references, regulatory_notes.

LOAN_DECISION_READY:

Decision Agent to Pricing Agent. Adds recommendation (APPROVE/DENY/COUNTER_OFFER), approved_amount, approved_tenure_months, decision_basis (which rule triggered), confidence (0-100%), compliance_notes (full audit note for regulatory records).

PRICING_TERMS:

Pricing Agent to Communication Agent. Adds interest_rate (percentage), processing_fee_pct, final_emi (monthly amount), prepayment_penalty (structure), insurance_premium (if applicable).

FORMAL_LETTER_READY:

Communication Agent to Human Gate (Streamlit). Adds letter_type (SANCTION_LETTER/REGRET_LETTER), letter_body (complete, formatted formal letter text ready for officer review and applicant dispatch), recommendation (carried forward).

INTAKE_ERROR:

Intake Agent to Human/System. Posted when input validation fails. Contains error_code and error_detail explaining what data was missing or invalid. HALTS the pipeline immediately — no subsequent agents are triggered.